Calculus II

Tutorial Problems #5 17–21 Mar 2025

Topics:

- Extreme values and saddle points
- Optimization problems

In Problems 1–2, find all the local maxima, local minima, and saddle points of the functions.

Problem 1. $f(x, y) = \sqrt{56x^2 - 8y^2 - 16x - 31} + 1 - 8x$.

Problem 2. $f(x, y) = e^{-y}(x^2 + y^2)$.

In Problems 3–5, find the absolute maxima and minima of the functions on the given domains.

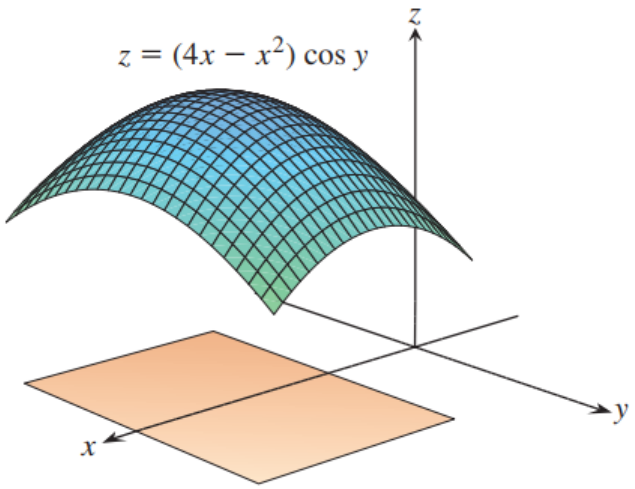
Problem 3. $f(x, y) = 2x^2 - 4x + y^2 - 4y + 1$ on the closed triangular plate bounded by the lines $x = 0$, $y = 2$, $y = 2x$ in the first quadrant.

Problem 4. $T(x, y) = x^2 + xy + y^2 - 6x + 2$ on the rectangular plate $0 \le x \le 5$, $-3 \le y \le 0$.

Problem 5. Find the absolute maxima and minima of the function on the given domain.

$$f(x, y) = (4x - x^2)\cos y$$

on the rectangular plate $1 \le x \le 3$, $-\pi/4 \le y \le \pi/4$ (see accompanying figure) .



Problem 6. *Temperatures* A flat circular plate has the shape of the region $x^2+y^2 \leq 1$. The plate, including the boundary where $x^2+y^2 = 1$, is heated so that the temperature at the point (x, y) is

$$T(x, y) = x^2 + 2y^2 - x.$$

Find the temperatures at the hottest and coldest points on the plate.

Problem 7. *Extrema on an ellipse* Find the points on the ellipse $x^2 + 2y^2 = 1$ where $f(x, y) = xy$ has its extreme values.

Problem 8. *Constrained minimum* Find the points on the curve $xy^2 = 54$ nearest the origin.

Problem 9. *Minimum surface area with fixed volume* Find the dimensions of the closed right circular cylindrical can of smallest surface area whose volume is $16\pi\, cm^3$.

Problem 10. *Rectangle of greatest area in an ellipse* Use the method of Lagrange multipliers to find the dimensions of the rectangle of greatest area that can be inscribed in the ellipse $x^2/16 + y^2/9 = 1$ with sides parallel to the coordinate axes.

END

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Answers to selected problems

1. $\left(\frac{1}{2}, \frac{3}{4}\right)$
3. max $= 1$, min $= -5$
4. max $= 11$, min $= -10$
5. max $= 4$, min $= \frac{3\sqrt{2}}{2}$
6. max $= \frac{9}{4}$, min $= 0$.
7. $(\frac{\sqrt{2}}{2}, \frac{1}{2})$, $((-\frac{\sqrt{2}}{2}, -\frac{1}{2}))$
8. $(3, \pm 3\sqrt{2})$
9. $A = 24\pi$ (when $r = 2$ and $h = 4$)
10. $A = (2\sqrt{2}, \frac{3\sqrt{2}}{2})$.